Neural means and kernel corrections for operator learning

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Abstract

Kernel regression can serve two roles after neural training: fitting prediction residuals or reading out learned features. We compare these roles on structural-mechanics and OCO-2 benchmarks, repeating the main experiments at ten seeds. On structural mechanics, a corrected six-member stack reaches $4.546 \pm 0.003\%$ mean relative test error, close to PARA-Net's published 4.55%. With 1250 labels, the pipeline reaches $5.433 \pm 0.093\%$ against a published 6.49%. An analysis of sixty trained predictors finds little remaining scope for global convex averaging: an empirical RMS lower bound of 4.814% is close to the hindsight optimum of 4.876%. On the O2 band, Matérn regression has approximately 40% reduced error on raw inputs and 4.1% on frozen neural features. Coordinatewise selection improves on the source emulator's stored predictions in the ten-seed means under both retained-representation criteria on all three bands. This comparison does not evaluate full-spectrum or retrieval accuracy. Ensemble identities and conditional native-space bounds delimit the interpretation of the results. For uncertainty quantification, a constant-radius conformal set is narrower on average than a set scaled by the kernel power function.

Keywords: operator learning, kernel ridge regression, residual correction, ensembles, conformal prediction

1 Introduction

A trained neural network leaves two natural places for a kernel regression: on the residual of its prediction, or on the representation learned by its hidden layers. Both constructions are well established, but their usefulness depends on the problem. We examine that dependence on two public operator-learning benchmarks, using the same basic components and repeating the main experiments at ten seeds.[1]

Operator learning approximates solution maps of partial differential equations and physical forward models. Neural operators learn representations suited to these maps (Kovachki et al., 2023; Lu et al., 2021; Li et al., 2021); kernel and Gaussian-process methods offer exact finite-design solves and conditional error bounds (Batlle et al., 2024; Mora et al., 2025; Owhadi and Scovel, 2019). Neural-mean Gaussian processes (Mora et al., 2025), deep kernel learning (Wilson et al., 2016; Calandra et al., 2016), and post-hoc feature-space residual models (Vendrell-Gallart et al., 2026) combine these approaches in different ways. Our contribution is a controlled empirical study of where the kernel helps, how the members' residuals limit gains from averaging, and how the evaluation metric affects the final

1. Code, data pointers, and per-run summaries are at `https://github.com/yspennstate/neural-means-kernel-corrections`. The online supplement is available there as `paper/supplement.pdf` and contains the full proofs.

Table 1: The two problems studied in depth. OCO-2 rows compare against the Gaussian-process emulator of Lamminpää et al. (2025), scored from its own stored predictions on the same 2000 test points, under the two retained-representation rescoring criteria defined here (Section 5); ours are mean ± standard deviation over ten seeds per band. Structural-mechanics rows quote the reference mean relative L^2 test error at each training size (de Hoop et al., 2022; Mora et al., 2025), reported without uncertainties in the original work; ours are mean ± standard deviation over ten seeds. Section 4.4 discusses the uncertainty of the 20000-sample comparison.

Problem	Metric	Reference result	This work
OCO-2, O2 band	reduced	16.89%	$4.12 \pm 0.05\%$
	radiance	0.0448%	$0.0294 \pm 0.0020\%$
OCO-2, weak CO_2	reduced	24.06%	$16.28 \pm 0.06\%$
	radiance	0.0599%	$0.0507 \pm 0.0052\%$
OCO-2, strong CO_2	reduced	16.14%	$8.08 \pm 0.01\%$
	radiance	0.1147%	$0.0584 \pm 0.0041\%$
Structural mechanics, 20000 samples	relative L^2	4.55% (PARA-Net)	$4.546 \pm 0.003\%$
Structural mechanics, 1250 samples	relative L^2	6.49% (FNO-mean GP)	$5.433 \pm 0.093\%$

combination. The supporting theory makes these comparisons precise without proposing a new hybrid construction.

On structural mechanics, a Matérn regression corrects the residual of a stacked neural mean. The six-member pipeline reaches $4.546 \pm 0.003\%$ mean relative test error, close to the published PARA-Net score of 4.55% (Table 1). The published score comes without predictions or retraining variability, so the two numbers are compared as point values. With a budget of 1250 labels, the pipeline reaches $5.433 \pm 0.093\%$ against a published 6.49%.

The structural-mechanics experiments also show why adding predictors yields diminishing returns. Residuals from different architectures are strongly aligned, and the hindsight-optimal global convex combination of sixty trained members has empirical root-mean-square relative error 4.876%. A lower bound from their within- and between-architecture second moments is 4.814% (Theorem 6.6). The narrow gap describes the remaining scope for global convex averaging in this pool. The bound concerns global convex weights within this pool; the per-pixel stacks and kernel corrections of the full pipeline lie outside it. The common residual may reflect shared approximation bias, numerical properties of the deterministic target, or both.

On OCO-2, the kernel's useful role changes. On the O2 band, the raw-input Matérn regression has approximately 40% reduced error, the flat-loss network 4.4%, and the same kernel on its frozen features 4.1%, under a common kernel grid and solve routine. The learned representation makes a substantial difference to the fitted kernel. Finite-design RKHS norms and power functions describe aspects of that difference, while the pullback identity supplies the relevant function-space interpretation; it does not predict a gain over raw inputs. A per-coordinate combination improves on the source emulator's stored predictions in both mean rescoring criteria on all three bands, with the weak-CO_2 radiance comparison winning at nine of ten seeds. These criteria concern the retained PCA representation, not full-spectrum instrument-noise acceptance or atmospheric retrieval.

Seed replication measures variability within the specified procedures, and paired evaluation measures differences on a common test block. The theoretical identities describe global convex ensembles exactly; the capacity and RKHS bounds hold under the assumptions stated with each result. Split-conformal prediction gives marginal coverage under its exchangeability conditions, and the coverage measurements here are made on the public test blocks. Among the calibrated sets examined, the constant-radius set is narrower on average than the power-function-scaled set.

Section 2 describes the data and comparison protocols. Section 3 specifies the implementation. Sections 4–5 present the experiments. Section 6 proves the block lower bound, the sharp native-space bound, and the minimax recovery result, and derives the correction-label mismatch term. The supplement contains the remaining proofs, implementation details, and additional measurements. Section 7 considers the implications.

2 Benchmark, protocol, and related work

2.1 Problem and data

The dataset originates in the cost-accuracy study of de Hoop et al. (2022) and is the one distributed with Batlle et al. (2024) and Mora et al. (2025). The domain $D = (0,1)^2$ is the unit cell of a fibre-reinforced composite: a cylindrical fibre of linear elastic material at the centre, embedded in an incompressible Rivlin–Saunders (hyperelastic) matrix, so the constitutive law is heterogeneous, with a material interface across which the stress jumps. The displacement field w satisfies $\nabla \cdot \sigma = 0$ under that law, with the bottom edge clamped and a prescribed tension traction $\bar{t}$ on the top edge $\Gamma_t = [0,1] \times \{1\}$. The quantity of interest is the von Mises stress field σ_v on D. The learning task is the map $G : \bar{t} \mapsto \sigma_v$. Loads are drawn from the Gaussian field $\mathcal{GP}\big(100,\, 400^2(-\Delta + 3^2 I)^{-1}\big)$ with homogeneous Neumann boundary conditions for the Laplacian on spatial-mean-zero fluctuations. In the source study the solver is a mesh of 189 quadratic quadrilateral elements (the NNFEM library) with the stress evaluated at its Gaussian quadrature points and transferred to a regular 41×41 grid by radial basis function interpolation; the distributed arrays are those grid values, with the load sampled at 41 points. The interface, the quadrature-to-grid transfer and the finite element discretization are all part of what a surrogate is asked to reproduce. The distributed file contains 40000 input/output pairs; the input array stores the load broadcast along the second grid coordinate, which we verified is an exact copy (maximal deviation 0 across all 40000 samples), so all our methods consume the load as a vector in $\mathbb{R}^{41}$.

Errors are mean relative L^2 errors over the test set,

$$\frac{1}{N_{\text{test}}} \sum_n \frac{\|\widehat{G}(u_n) - G(u_n)\|_2}{\|G(u_n)\|_2},$$

computed on grid values in double precision using unweighted vector norms. Published comparisons retain their source values; Batlle et al. (2024), for example, use trapezoidal quadrature for their discretized L^2 norm, so errors quoted from different studies agree in definition only up to such conventions.

2.2 Protocols and published results

Two regimes appear in the literature, and we follow both. In the *high-data* protocol the first 20000 samples are available for training and the last 20000 form the test set. The neural-network results of de Hoop et al. (2022), collected in Table 3 of Batlle et al. (2024), include, at 20000 training samples, 4.55% (PARA-Net), 4.67% (PCA-Net), 4.76% (FNO) and 5.20% (DeepONet), and Batlle et al. (2024) report 5.18% for their optimal-recovery kernel method (Matérn/rational quadratic; 27.11% for the linear kernel) on the same task. Within the 20000-sample budget we hold out the first 1000 samples of a fixed permutation for model selection and stacking, and train on the remaining 19000, so no method of ours sees more than the 20000 training samples used by the baselines. In the *low-data* protocol of Mora et al. (2025), 1250 samples are available and the same 20000-sample test set is used; their table gives 8.70% (DeepONet), 6.62% (FNO), 6.95% (the kernel method of Batlle et al. 2024), 6.74% (their zero-mean GP), 7.12% (DeepONet-mean GP) and 6.49% (FNO-mean GP). In this regime we carve the validation split (250 samples) out of the 1250, so training uses 1000 samples. Fitting and selection within each low-data run use those 1250 labels.

2.3 The rest of the suite, and a check of the mirror symmetry

Supplement S4 reports a survey of the other operator-learning problems and a check of reflection symmetry. The survey uses a largely fixed recipe with unequal splits and limited seed replication and is included for context. For structural mechanics, reflected near-neighbor pairs, the reflection axis, and empirical load moments are consistent with the symmetry used by Proposition S1.1. These finite-data checks support the modeling assumption.

2.4 Related work

Our comparisons draw on neural operators and kernel methods for learning maps between function spaces. DeepONet (Lu et al., 2021), the Fourier neural operator (Li et al., 2021), and the PCA-Net and PARA-Net models in de Hoop et al. (2022) provide the structural-mechanics reference results; Kovachki et al. (2023) reviews the broader framework. The optimal-recovery kernel method of Batlle et al. (2024) provides a competitive alternative with native-space error bounds. Data-adapted kernels (Darcy et al., 2023) extend this approach beyond a fixed input metric.

Neural means with Gaussian-process corrections are already established. Mora et al. (2025) optimize neural-operator means through a Gaussian-process likelihood, keeping the kernel parameters fixed in their neural-mean experiments. We fit the mean first, select the kernel on validation, and solve the correction at all 19000 training points. The underlying discrepancy construction also appears in multifidelity modeling (Kennedy and O'Hagan, 2000) and reduced-order-model error surrogates (Drohmann and Carlberg, 2015). A frozen trained mean gives ordinary Gaussian conditioning on its residuals; this differs from universal kriging when the trend coefficients are estimated jointly from the same observations.

Kernels on learned representations are likewise not new. Deep kernel learning (Wilson et al., 2016; Calandra et al., 2016) trains through a Gaussian process on network features. REEF-GP (Vendrell-Gallart et al., 2026) fits a Gaussian process post hoc to a neural operator's residuals in its feature space. Our contribution here is an empirical comparison of raw inputs and frozen features under the same kernel family, search grid, tuning subsample, and full refit, with a linear readout control. This matches the kernel computation, not the extra cost of learning features. Separating network training from kernel selection also avoids joint optimization, whose overfitting behavior is examined by Ober et al. (2021).

Other uncertainty constructions induce a Gaussian process from a neural operator (Kumar et al., 2024) or from its linearization (Magnani et al., 2025). Ma et al. (2024) apply split conformal calibration to operator predictions. We use the same rank-based calibration principle for an L^2 ball around the whole output, a different coverage event from their spatial-coverage formulation. Our reported coverage is measured on the public test block.

Two further connections help interpret the experiments. Kernel flows (Owhadi and Yoo, 2019) learn kernels by half-sample cross-validation; Yoo and Owhadi (2021) use this loss to regularize network features, as analyzed in Proposition S1.2. Applications include radiative-transfer emulation (Lamminpää et al., 2025) and satellite inference of storm structure (Prasanth et al., 2021). Effective dimension links the kernel solve to classical Gram-spectrum and random-matrix results (Koltchinskii and Giné, 2000; Marčenko and Pastur, 1967; Baik et al., 2005) and to kernel-ridge generalization theory (Caponnetto and De Vito, 2007). We use these results as reference models.

3 Method

The pipeline has three stages: neural means, stacking, and kernel corrections. All components operate on the load vector $u \in \mathbb{R}^{41}$ (standardized by training statistics) and produce stress fields in $\mathbb{R}^{41\times 41}$. Neural members use either mean relative L^2 loss in original units or normalized mean-squared loss, as identified in Table 4. The normalized-MSE MLP is the stronger of the two plain MLPs in these experiments.

3.1 Neural means

Residual MLP. The primary mean is a residual multilayer perceptron: three or five residual SiLU blocks of width 1024–1536 mapping $\mathbb{R}^{41} \to \mathbb{R}^{1681}$ directly. This is essentially PARA-Net (de Hoop et al., 2022) with a modern training recipe, and it also supplies the penultimate features used by the feature-space kernel stage.

Kernel-conditioned refiner. The second mean is the same residual network given an extra input channel: the kernel method's predicted stress field for the same load, concatenated with the load. The network predicts the full stress field from these two inputs; it has no explicit residual-output constraint. Its test error is close to that of the normalized-MSE network (Table 4). During training the refiner reads four-fold kernel predictions: the kernel weights are fitted excluding the held-out fold, with targets centered by the mean over the full fitting pool. At query time the refiner receives the full-data kernel prediction. Section 4.8 reports a paired experiment with fold-local target centering.

Other instances. The mean slot is not tied to a particular architecture. We also use a Fourier neural operator (Li et al., 2021) that consumes the broadcast load as a field and an MSE-trained variant of the MLP; both are drop-in means, both enter the ensemble of Section 4.5, and their pairwise error correlations are one of the measurements of this paper. A UNet on the same field representation is the sixth member of the second campaign (Section 4.1). We additionally implement an encoder–decoder transformer (Vaswani et al., 2017; Dosovitskiy et al., 2021) that tokenizes the 41 load samples and decodes the field by cross-attention at the grid points; it is not part of the reported ensemble (Supplement S9).

All means are trained with AdamW and a cosine schedule, batches of 128–256, for up to a few hundred epochs (Supplement S9 lists every hyperparameter). The neural trainers reflect batches with probability 1/2 (load reversed, target field reflected along the first grid coordinate). Direct neural predictors use $\frac{1}{2}(f(u) + Tf(Su))$ at evaluation. For the refiner, write $F(u, h)$ for its prediction from a load and a supplied kernel field. Its implemented average is

$$\tfrac{1}{2}\big(F(u, h(u)) + TF(Su, Th(u))\big).$$

The reflected branch permutes the supplied field rather than recomputing $h(Su)$. This equals reflection averaging of the composite predictor $f(u) = F(u, h(u))$ if $h(Su) = Th(u)$; the fitted kernel channel is not constrained to satisfy this identity. Proposition S1.1 gives the population-risk guarantee for the full-map average under exact target equivariance and distribution invariance, with the channel condition as an additional hypothesis for the refiner. The kernel-only member uses neither neural augmentation nor this prediction average. Training augmentation is a separate intervention from prediction averaging. Optionally, the training loss includes a regularized kernel-flow term computed from the batch with a Gaussian kernel on penultimate features and a learned bandwidth multiplier. The implementation adds $10^{-3}I$ to the half-batch Gram matrix and treats the batch median distance as constant in differentiation. Section S1.2 distinguishes this ablation from the ideal interpolation loss analyzed there.

3.2 Stacking

With M trained models $f_1, \dots, f_M$ we form the convex combination $m(u) = \sum_m w_m f_m(u)$, $w \in \Delta^{M-1}$, minimizing the validation relative error; the deployed weights start uniform and are found by a random hill climb in logit space (600 Gaussian proposals of step 0.3, decaying by 0.3% per step) on half of the 1000 held-out samples (250 in the low-data protocol), the other half judging the per-pixel variant against them. The simplex minimizer of the measured second-moment matrix of Proposition 6.1 is used in the analyses of Section 4.6, not in the deployed stack. A per-pixel variant allows the weights (plus an intercept) to vary over the grid, ridge-fitted on half the validation split and accepted only when it beats the global weights on the other half; it is the variant the final pipeline uses. Validation-selected stacking (Wolpert, 1992) is an additional fitted stage; Proposition S1.3, proved in Supplement S8, bounds its selection cost for independent candidates.

3.3 Kernel corrections

Let M_{tr} be the training-row mean. It combines foldwise kernel predictions with in-sample predictions from fitted neural members; the refiner is evaluated using its training kernel channel. Its residual labels are $R = Y - M_{\text{tr}}$. At query time the mean is m_{full}. These need not agree on training inputs. Equation (5) gives the resulting deterministic mismatch term.

The stacked mean is then corrected by kernel ridge regression of its residuals. Let $X = (u_1, \dots, u_n)$ be the training loads (standardized coordinatewise), $R \in \mathbb{R}^{n \times 1681}$ the matrix of training residuals $R_i = v_i - (M_{\text{tr}})_i$, and k a Matérn-5/2 kernel $k(u, u') = \kappa\big(\|u - u'\|/s\big)$.

The correction is

$$\widehat{r}(u) = k(u, X)\,\alpha, \qquad \alpha = (K + n\lambda I)^{-1}R,$$

with (s, λ) chosen on the validation split from a small grid (s as a multiple of the median pairwise distance, $\lambda \in \{10^{-6}, 10^{-5}, 10^{-3}\}$; the grid is searched on an 8000-sample subsample and the chosen pair is refit on all of X). The corrected surrogate is $m_{\text{full}} + \widehat{r}$. A second corrective stage repeats the construction with the kernel evaluated on deep features, the concatenated penultimate activations of the ensemble members (reflection-averaged, standardized), fitted to the training-row residuals after the first correction; it contributes a smaller improvement and is included when it helps on validation. On OCO-2 (Section 5) the kernel stage takes a different form: a direct-target kernel ridge regression on the network's frozen last-layer features, tuned and refit by the same routine, in place of a residual correction of the network; the two uses are named separately where they appear. At $n = 19000$ the exact solve is a single Cholesky factorization of a 19000×19000 matrix, under half a minute in double precision on a CPU host (factorization alone; Section S6.4). Prediction evaluates the kernel against the training inputs and multiplies by α; no inducing points, preconditioners, or stochastic solvers are involved. The Gaussian process reading of the same formulas supplies the posterior standard deviation $P_\lambda(u)$ of (4) by a triangular solve per evaluation batch, which is the design factor in the residual bound of Theorem 6.9 and the uncertainty signal studied in Section 4.7.

The construction uses the benchmark's reflection symmetry but no other specific property of its governing equation. It can therefore be implemented for other low-dimensional input parametrizations, with symmetry averaging where appropriate. The ablations measure the contribution of the feature stage, stacking, and residual correction separately.

4 Results on the structural-mechanics benchmark

4.1 Seeds and replication

The high-data experiments use ten seeds. Each seed redraws the validation carve from the 20000-row training block and reinitializes the neural members; the 20000-row public test block stays fixed. The first campaign uses a kernel predictor, a metric-loss MLP, a normalized-MSE MLP, a kernel-conditioned refiner, and an FNO trained on a partial schedule whose retained checkpoints lie between epochs 10 and 70. The second campaign trains the FNO through a complete 100-epoch schedule and adds a UNet. It reproduces the other members to the printed precision. The low-data experiment also has ten seeds but keeps its 1000/250 training/validation split fixed.

The comparisons below use the pooled-centering procedure of Section 3; Section 4.8 reports a paired comparison with fold-local target centering. Model-level selection uses validation.

4.2 Main comparison

Table 2 quotes published high-data scores and the two pipelines. Our kernel reproduces the published kernel score approximately (5.197% versus 5.18%), which supports broad protocol agreement. The six-member pipeline reaches $4.546 \pm 0.003\%$, against PARA-Net's published 4.55%. The low-data result is $5.433 \pm 0.093\%$ versus a published 6.49% (Table 3).

Table 2: Structural mechanics, high-data protocol (20000 training samples, 20000 test). Mean relative L^2 test error. Rows above the rule are published results on the same data, quoted without uncertainties because none were reported. Rows below are ours: mean ± standard deviation over ten seeds. The four-member row is the same pipeline with the FNO removed, kept because the FNO's admission is decided on validation (Section 4.3); the last two rows are the second campaign of Section 4.1, with the FNO trained to the end of its schedule and a UNet added. TTA denotes reflection averaging at test time.

Method	Error (%)	Parameters
DeepONet (de Hoop et al., 2022)	5.20	–
FNO (de Hoop et al., 2022)	4.76	–
PCA-Net (de Hoop et al., 2022)	4.67	–
PARA-Net (de Hoop et al., 2022)	4.55	–
Optimal-recovery kernel (Batlle et al., 2024)	5.18	–
Kernel ridge on loads (ours)	5.197 ± 0.003	–
Residual MLP + reflection TTA	4.836 ± 0.018	4.9M
+ affine stack over five members	4.595 ± 0.008	22.9M
+ residual kernel correction	4.572 ± 0.010	22.9M
the same pipeline without the FNO	4.611 ± 0.005	16.5M
+ complete FNO schedule and UNet, affine stack over six	4.567 ± 0.004	27.3M
+ residual kernel correction	4.546 ± 0.003	27.3M

Table 3: Low-data protocol (1250 training samples, 20000 test). Published rows from Mora et al. (2025). Our pipeline sees only the 1250 labels (250 of them used as the validation split). The pipeline row is mean ± standard deviation over ten seeds. Which stage the validation split selects varies with the seed — the input-space kernel correction at six seeds, its two-stage form at three, the feature-space one at the remaining seed — so the row is the error of the selected pipeline, not of a fixed final stage. The kernel-ridge row is the multiscale construction of Supplement S9, which has no random component under this protocol and therefore no seed spread.

Method	Error (%)
DeepONet	8.70
GP, optimal recovery (Batlle et al., 2024)	6.95
Zero-mean GP (Mora et al., 2025)	6.74
FNO	6.62
FNO-mean GP (Mora et al., 2025)	6.49
Kernel ridge on loads (ours, multiscale)	6.66
Full pipeline (ours, ten seeds)	5.433 ± 0.093

Table 4: Members and stages under the high-data protocol, mean $\pm$ standard deviation over ten seeds. Neural-member errors use reflection averaging; the refiner also uses reflected training examples. The standalone kernel-ridge member uses neither reflection augmentation nor prediction averaging. "Fitted" weights are solved on the validation split and read out once on test; equal weights involve no fitting. The lower block is the second campaign of Section 4.1; its kernel ridge and MLP-family members coincide with the upper block's to the printed precision and are not repeated. The kernel ridge member and the correction each store a 19000×1681 coefficient matrix (31.9 million numbers, 256 MB in double precision) beside the networks' weights; the accuracy–cost–storage record is in Section S6.4.

Member or stage	Family / loss	Test error (%)
Kernel ridge on loads	Matérn-5/2	5.197 ± 0.003
Residual MLP	MLP, metric loss	4.836 ± 0.018
Residual MLP	MLP, normalized MSE	4.712 ± 0.006
Kernel-conditioned refiner	MLP on $(u, \mathrm{KRR}(u))$	4.730 ± 0.006
Fourier neural operator	spectral, metric loss, early-stopped	4.754 ± 0.031
Equal-weight ensemble	no fitting	4.648 ± 0.005
Fitted-weight ensemble	solved on validation	4.627 ± 0.006
Affine stack (per-pixel)	selected on a held-out half	4.595 ± 0.008
+ residual kernel correction	–	4.572 ± 0.010
Fourier neural operator	spectral, metric loss, complete schedule	4.682 ± 0.009
UNet	convolutional, metric loss	4.740 ± 0.030
Equal-weight ensemble, six members	no fitting	4.626 ± 0.003
Fitted-weight ensemble, six members	solved on validation	4.602 ± 0.003
Affine stack (per-pixel), six members	selected on a held-out half	4.567 ± 0.004
+ residual kernel correction	–	4.546 ± 0.003

4.3 Members and the stack, across seeds

Table 4 gives the stage-by-stage accounting. In the first campaign, the normalized-MSE network has the lowest single-member test mean (4.712%), followed by the refiner and the early-stopped FNO; the metric-loss MLP is less accurate. Adding the spectral member to the four-member pipeline reduces mean error by 0.039 ± 0.011 percentage points, with validation preferring its admission at all ten seeds. The second campaign's completed FNO schedule improves its individual mean by 0.072 points, but the five-member corrected pipeline changes from 4.572% to 4.578%, a paired difference of 0.006 ± 0.013 points. Increased alignment with the other members is consistent with that lack of ensemble gain.

The UNet ranks fourth of six by mean single-member test error, yet adding it reduces the corrected pipeline's error by 0.032 ± 0.004 points. In the first campaign, equal weighting, fitted global weights, per-pixel affine weights, and kernel correction successively reduce mean error by 0.064 ± 0.007, 0.021 ± 0.003, 0.032 ± 0.005, and 0.024 ± 0.005 points; each improvement occurs at ten of ten seeds. These are paired ablations within this trained pool.

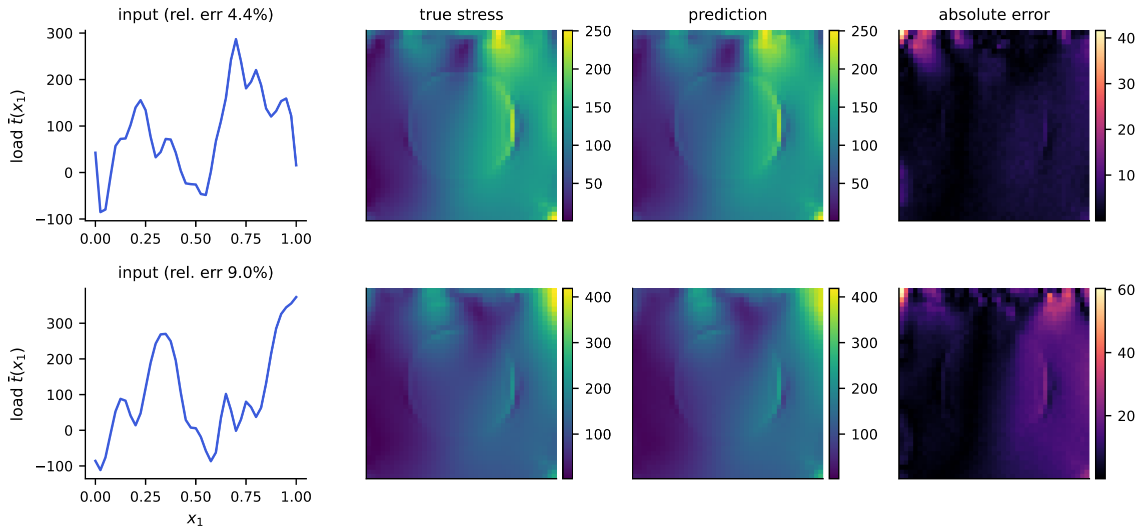

Figure 1: Example predictions of the full pipeline. Top: a median-error test case; bottom: a case at the 98th error percentile. Columns: input load, true von Mises stress, prediction, and pointwise absolute error (note the smaller scale). Errors concentrate in the high-traction corner, as also reported by Mora et al. (2025).

4.4 Which uncertainty is the binding one

The seed standard deviations measure training and selection variability on a fixed test block. Separately, the observed per-case standard deviation 0.0169 gives a standard error of about 0.012 percentage points for the mean over 20000 i.i.d. cases, conditional on a fixed predictor. These are different quantities. The published comparator comes without per-case predictions or retraining variability, so no significance or equivalence test against PARA-Net is available; within-study comparisons use paired per-case differences, and their seed spreads describe sensitivity to retraining.

4.5 Architectural diversity and the shared residual

The centered residual correlations range from 0.82 to 0.98 across the six-member campaign, with mean 0.918 ± 0.005. These correlations describe alignment. The exact ensemble identity uses the uncentered normalized-residual second-moment matrix S, which is a distinct quantity.

For each seed, the analysis forms S on validation and minimizes $w^\top S w$ over the simplex. The resulting validation RMS errors are $4.940 \pm 0.057\%$ for five members and $4.920 \pm 0.057\%$ for six. The factors 0.938 ± 0.003 and 0.938 ± 0.003 compare the mean and RMS errors of the same fitted mixture on that same validation set. They are dispersion diagnostics. For six members, the test-mean/validation-RMS ratio is approximately 0.936 ± 0.011; this ratio combines metric dispersion and transfer between samples.

The equicorrelation references 4.922% and 4.908%, formed from average centered correlations and member RMS errors, are descriptive approximations. The rigorous entrywise lower bounds of Corollary 6.3 are computed directly from the measured matrix in Section 6.2.

The across-member mean residual carries about 90% of average residual energy on validation and concentrates near the corners of the loaded edge. It has more high-frequency energy than the target fields, and leading residual directions align across architectures (Supplement S6). These observations establish shared structured error in the trained pool without identifying its cause: common surrogate bias and finite-element or interpolation effects are both possible, and since the target is a deterministic numerical map, solver discrepancy relative to a continuum solution is not by itself irreducible error for learning the numerical labels.

Learning curves show modest additional gains over the tested data range: the normalized-MSE model moves from $4.799 \pm 0.003\%$ at 8000 fitting rows to $4.712 \pm 0.004\%$ at 19000. Supplement S6 contains the learning curves, residual plots, ablations, and cost measurements.

4.6 Seeds against architectures

The sixty-member analysis uses ten seeds of six model configurations, treating these configurations as blocks. The public test block is split into 1000 calibration cases for fitted weights and 19000 evaluation cases; hindsight optima are explicitly computed on evaluation. The RMS-optimal global convex mixture records 4.876% RMS error and 4.581% mean error. The latter is the mean error of the RMS-optimal weights, not a minimum of the mean-error objective.

Theorem 6.6, applied to the evaluation matrix and its smallest within- and between-block entries, gives 4.814% RMS as an empirical lower bound for every global convex combination of these sixty predictors on these cases. The gap to the recorded optimum is approximately 0.063 percentage points. The bound concerns global convex combinations of these sixty predictors; per-pixel affine and kernel-corrected estimators lie outside its class.

The deployed six-member corrected pipeline has mean error $4.543 \pm 0.003\%$ on the same evaluation cases. This is below the mean error of the RMS-optimal convex mixture. The pipeline fitted to the six ten-seed mean predictors reaches 4.533%, a gain outside the theorem's estimator class. Table 5 lists the fitted-stack comparisons in both metrics.

4.7 Uncertainty quantification

The single-mean power-function bound and its correction-label mismatch extension appear in Section 6.4. Section 4.8 measures the propagated mismatch; the residual RKHS norm itself is not observed, so the bound is used as a structural statement rather than as a numerical certificate. Fitted-function norms are computable diagnostics: the squared-norm ratio is about 4900 before normalizing target energy and about 8.8 after it (Table S6.2).

The power function has weak correlation with the absolute error (0.055) and negative correlation with relative error, while correlating with output norm (0.65). Disagreement is similarly weak as an error indicator (0.074 ± 0.006 against absolute error). Agreement among predictors cannot reveal an error component shared identically by all of them, and neither scale ranks the error well here.

For the six-member pipeline, conformal calibration on 1000 cases and evaluation on the disjoint 19000 give $89.9 \pm 0.8\%$ coverage at nominal 90% and $95.0 \pm 0.6\%$ at nominal 95%. Under the frozen-predictor i.i.d. continuous-score assumptions of Remark S3.2, the

Table 5: Sixty predictors, ten seeds of six architectures, on the evaluation subset of the test block (19000 cases). Mean relative L^2 error, and the root mean square where the theory speaks in that metric. Convex weights are fitted on the disjoint calibration subset (1000 cases) unless marked hindsight, where they are chosen on the evaluation subset itself. Rows with $\pm$ are means over the ten seeds.

Pool and weights	Error (%)	RMS (%)
FNO, one seed	4.679	4.967
FNO, ten seeds, equal weights	4.649	4.940
Six architectures, one seed, equal weights	4.623 ± 0.003	4.916 ± 0.003
Six architectures, one seed, convex, fitted	4.600 ± 0.003	–
Sixty, equal weights	4.611	4.905
Sixty, convex, fitted	4.586	4.880
Sixty, convex, hindsight	4.581	4.876
Lower bound of Theorem 6.6 on the sixty	–	4.814
Six ten-seed means, per-pixel affine, fitted	4.556	–
Sixty, per-pixel affine, leave-one-out ridge, fitted	4.547	–
Six means and the learned kernel, per-pixel affine, leave-one-out ridge	4.546	–
Six ten-seed means, per-pixel affine and kernel correction, fitted	4.533	–
Deployed six-member pipeline, one seed	4.543 ± 0.003	–

corresponding central 95% reference intervals are $[88.0\%, 91.8\%]$ and $[93.5\%, 96.3\%]$; all ten seeds lie inside each.

A constant-radius conformal set gives $90.4 \pm 0.8\%$ and $95.4 \pm 0.5\%$ coverage and is narrower on average: the power-function set has mean radius 1.11 times as large. The power-function set also varies markedly across deciles, from 69% to 99.9% coverage at seed zero. Thus marginal coverage is compatible with poor local coverage and weak error ranking. Supplement S6 includes all three scales and the five-member comparisons. The formal guarantee is stated with its hypotheses and proof pointer in Section 6.5.

4.8 Sensitivity to centering and correction labels

Two additional experiments examine the distinction between the training-row kernel predictions and the predictor evaluated at a new input. First, we rerun the six-member pipeline at the same ten seeds, changing only the target mean used in each out-of-fold kernel fit. Both arms retrain the kernel-conditioned refiner for 100 epochs and refit the stack and residual correction; the remaining members are held fixed. Pooled centering gives $4.5459 \pm 0.0029\%$ mean relative error and fold-local centering gives $4.5459 \pm 0.0029\%$. The paired difference, fold-local minus pooled, is 0.000011 ± 0.000110 percentage points. Figure 2 shows every seed, including the direction of the change. This comparison isolates the effect of target centering.

Second, we hold the stack weights and selected correction kernel fixed, evaluate the query-time mean on the training rows, and replace the correction labels by residuals of that mean. No network is retrained in this check. The resulting test error is $4.5460 \pm 0.0029\%$, a paired change of 0.000007 ± 0.000029 percentage points from the original correction. The propagated mismatch $\|c_\lambda(u)^\top \Delta\|_2 / \|G(u)\|_2$ from (5) averages $0.0148 \pm 0.0035\%$ across seeds

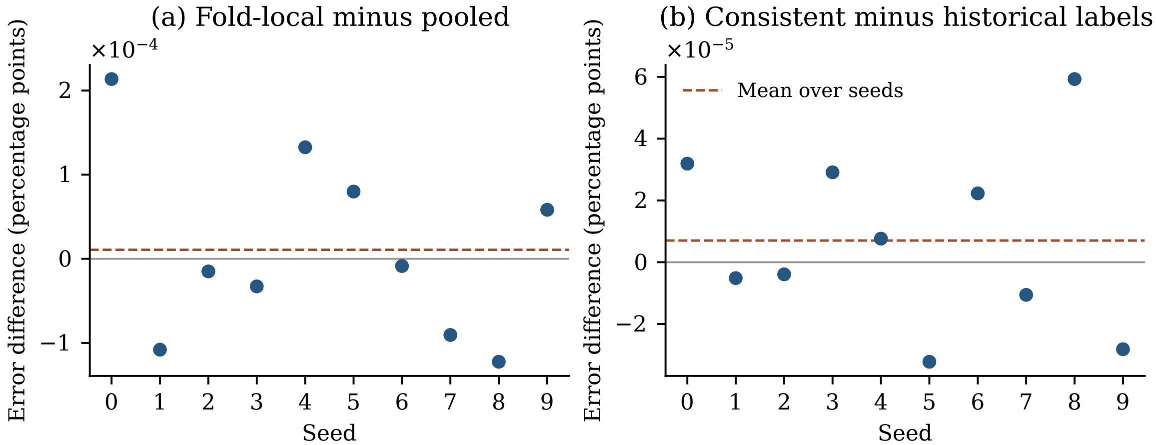

Figure 2: Paired changes in mean relative test error for the ten mechanics seeds. Left: fold-local minus pooled target centering, with downstream refiner training and fitting repeated in both arms. Right: correction labels consistent with the query-time mean minus the original labels, with the mean and kernel fixed. Positive values mean higher error after the change. The panels use separate vertical scales. All points use the same public test block; the dashed lines are seed means.

and cases. It measures the difference between two finite-design predictors. Supplement S11 gives the protocol, full stage comparisons, and calibration results.

5 The OCO-2 radiative-transfer emulator

The structural-mechanics benchmark put the neural mean and the kernel in the balanced regime. To see the other regime we apply the same components to the radiative-transfer emulation problem of Lamminpää et al. (2025): per spectral band of the OCO-2 instrument, learn the map from a reduced atmospheric state (20–24 dimensions after the dimension reduction of that paper) to the reduced radiance (40 PCA coefficients of the monochromatic spectrum). The OSF project (`u2t8a`) publishes a training pool of 20000 pairs and, separately, 2000 test states together with the kernel-flow emulator's own predictions on them; the released overlap check reports that the two sets share no point. The runs split the pool 18000/2000 into training and validation. Within each run, the validation split selects network checkpoints, kernel length scales and nuggets, and the coordinate winners below. The public test states are used for the reported evaluation. Section 5.4 reports a separate matched sensitivity experiment. The carve is taken from the source study's Sobol training design while the public test states are a separate draw, and the two differ: the flat network's mean relative error is 6.7% on the carve against 4.4% on the test block on O2, so every selection made on the carve is a selection under a shift between the two, not a same-distribution choice. The comparison is therefore computed on identical test points against the published emulator itself rather than against our reimplementation of it.

The two error criteria are defined here on the released reduced representation. Neither is the source study's instrument-noise or retrieval acceptance test. Let $z \in \mathbb{R}^{40}$ contain the standardized coefficients, let U have the retained orthonormal PCA columns, and put $D_z = \mathrm{diag}(s_z)$. The stored means and scales reconstruct

$$R(z) = \mu_y + U(\mu_z + D_z z), \qquad U^\top U = I. \tag{1}$$

The two reported mean relative errors are

$$E_{\mathrm{red}} = \frac{1}{N}\sum_{i=1}^{N} \frac{\|\hat{z}_i - z_i\|_2}{\|z_i\|_2}, \tag{2}$$

$$E_{\mathrm{rad}} = \frac{1}{N}\sum_{i=1}^{N} \frac{\|D_z(\hat{z}_i - z_i)\|_2}{\|R(z_i)\|_2}. \tag{3}$$

Orthogonality gives $\|R(\hat{z}) - R(z)\|_2 = \|D_z(\hat{z} - z)\|_2$; the reconstruction offsets cancel in the numerator but remain in the denominator. The radiance criterion compares two forty-component reconstructions, so it excludes PCA truncation error, the instrument line shape, and the measurement-noise model. Its numerator weights concentrate on the leading coefficients.

For fixed nonnegative weights ω_i, a_j, the squared objective $N^{-1}\sum_{i,j}\omega_i a_j(\hat{z}_{ij} - z_{ij})^2$ separates by coordinate. Ordinary coordinatewise RMSE selection minimizes this objective with $\omega_i = a_j = 1$. Squared relative coefficient error instead uses $\omega_i = \|z_i\|_2^{-2}$; squared relative radiance error uses $\omega_i = \|R(z_i)\|_2^{-2}$ and $a_j = s_{z,j}^2$. Fixed per-sample denominators therefore preserve separability for the squared objectives, although they can change the winning member. The square root in each summand of (2)–(3) couples coordinates, so coordinatewise selection need not minimize either reported mean norm, or both simultaneously.

Table 6 and Figure 3 show the principal comparisons. The tested kernel routine scores 40% on raw input, 30% after rescaling each input coordinate by its estimated sensitivity, and 4.1% on the trained network's features. Under this routine, learned features are more useful than the raw coordinates or the tested input rescaling. The deep kernel head is the strongest single model tested on the O2 reduced metric. Its margin over the network it reads out is $4.11 \pm 0.05\%$ against $4.43 \pm 0.05\%$, and the head has lower error at all ten seeds on each band.

A linear ridge regression refitted on the same frozen features (Table 7) is worse than the network's trained output layer on the flat-loss network at every band-seed. The Matérn head is better than both at every band-seed, with a margin over ridge of 0.60 ± 0.07 percentage points on O2 and about a quarter of a point on the other bands. Thus the linear-refit control does not reproduce the Matérn improvement. Adding the ridge readouts to coordinate selection changes the combination by at most 0.002 percentage points on any band.

Three details of the comparison belong next to these numbers. The combination row is produced by the unweighted selector over the three networks and their three feature-kernel heads. It takes each reduced coordinate from the model with the smallest validation root-mean-square error on that coordinate; we also ran a selector weighted by the inverse squared norm of each validation state, the exact optimizer over the coordinatewise candidate choices of the empirical squared relative error described by Corollary S2.3, and it chooses different winners at three seeds on O2 and four on WCO2 yet changes each band's mean reduced error by less than 0.001 percentage points, with unchanged radiance means at the reported

Table 6: OCO-2 O2 band, scored on the 2000 public test states, which are disjoint from the training pool; within-run selection used a validation split of the pool. Every fitted row is mean ± standard deviation over ten seeds; neural networks use a matched 250-epoch budget. The kernel-flow row is the emulator of Lamminpää et al. (2025), scored from its own stored predictions on fixed test points, and is identical at every seed. "Features" means the penultimate activations of the trained network; the kernel head is the same exact Matérn solve as everywhere else in this paper, and the two raw-input rows are fit by that same kernel-tuning routine and budget as the feature heads: the same scale grid $\{0.5, 1, 2, 4\}$ times the median distance, the same nugget grid $\{10^{-8}, 10^{-6}, 10^{-4}\}$, the same 6000-sample tuning subsample, and the winner refit on all 18000 training points. This matches the kernel stage, not total computation: feature heads additionally require neural-network training. The raw-input rows' selected settings vary from seed to seed; every feature head, by contrast, selected the largest multiplier (4) and the smallest nugget (10^{-8}) of that grid at all thirty band-seeds, so the feature-head numbers are tied to this search grid; Section 5.4 examines a wider one. The ARD radiance summary includes a bimodal selection outcome: configurations selected on reduced validation error give 0.150% radiance error at seven seeds and 0.329% at three.

model	reduced	radiance
Matérn kernel, raw input, one length scale	$40.57 \pm 0.10\%$	$0.233 \pm 0.003\%$
Matérn kernel, raw input, sensitivity-scaled (ARD)	$29.98 \pm 0.15\%$	$0.204 \pm 0.087\%$
kernel-flow emulator (Lamminpää et al., 2025)	16.89%	0.0448%
residual MLP, flat metric	$4.43 \pm 0.05\%$	$0.195 \pm 0.014\%$
residual MLP, weighted-coefficient loss	$49.32 \pm 0.40\%$	$0.0442 \pm 0.0044\%$
residual MLP, exact radiance loss	$59.02 \pm 0.76\%$	$0.0571 \pm 0.0057\%$
Matérn head on the flat network's features	$4.11 \pm 0.05\%$	$0.0793 \pm 0.0072\%$
Matérn head on the weighted network's features	$21.70 \pm 0.52\%$	$0.0319 \pm 0.0020\%$
Matérn head on the radiance network's features	$20.48 \pm 0.61\%$	$0.0379 \pm 0.0037\%$
per-coordinate combination	$\mathbf{4.12 \pm 0.05}\%$	$\mathbf{0.0294 \pm 0.0020}\%$

precision. On O2 the unweighted combination has mean reduced error 4.11510%, slightly above the flat-feature head's 4.11080%, while improving radiance error from 0.07927% to 0.02940%. It therefore does not dominate every individual model on both criteria. The uncertainty of the comparison against the emulator is paired: on the same 2000 test states, a case-resampling bootstrap of the difference between the combination and the emulator excludes zero at ten of ten seeds on both metrics on O2 and SCO2, and at ten of ten on WCO2's reduced metric and nine of ten on its radiance metric, the exception being the seed that loses by 0.0015 points; the seed standard deviations in Table 6 describe retraining variability, not that comparison. The comparison concerns the two criteria defined here on the released reduced representation (the coefficients, and the radiance reconstructed from the retained forty components) rather than the full-spectrum acceptance test of Lamminpää et al. (2025), with the networks at a 250-epoch budget and the emulator fitted on 20000 states against our 18000 plus a 2000-state validation carve. Instrument response, Jacobians and the retrieval itself are outside its scope. Learning the metric rather than setting it

Table 7: Readout control for the kernel heads, ten seeds per band, mean ± standard deviation over seeds. For the flat and weighted-coefficient networks the same frozen last-layer features are read out three ways: by the network's own trained output layer, by a ridge regression refit on those features with the penalty chosen on validation, and by the exact Matérn head. The rows come from the campaign script with the ridge rows added (`campaign/jpl_seeded.py`; body generated by `campaign/gen_oco_ridge_table.py`); its network and head rows reproduce Table 6 within the seed spread.

model	reduced	radiance
O2 band (10 seeds)		
network, flat loss	$4.43 \pm 0.05\%$	$0.195 \pm 0.014\%$
ridge readout on its frozen features	$4.71 \pm 0.07\%$	$0.228 \pm 0.014\%$
Matérn head on its frozen features	$4.11 \pm 0.05\%$	$0.0793 \pm 0.0072\%$
network, weighted-coefficient loss	$49.33 \pm 0.40\%$	$0.0439 \pm 0.0031\%$
ridge readout on its frozen features	$45.71 \pm 0.61\%$	$0.0521 \pm 0.0043\%$
Matérn head on its frozen features	$21.72 \pm 0.52\%$	$0.0316 \pm 0.0019\%$
WCO2 band (10 seeds)		
network, flat loss	$16.44 \pm 0.06\%$	$0.226 \pm 0.012\%$
ridge readout on its frozen features	$16.51 \pm 0.06\%$	$0.252 \pm 0.017\%$
Matérn head on its frozen features	$16.28 \pm 0.06\%$	$0.116 \pm 0.007\%$
network, weighted-coefficient loss	$62.57 \pm 0.56\%$	$0.0738 \pm 0.0057\%$
ridge readout on its frozen features	$61.32 \pm 0.43\%$	$0.0842 \pm 0.0067\%$
Matérn head on its frozen features	$48.67 \pm 0.45\%$	$0.0533 \pm 0.0054\%$
SCO2 band (10 seeds)		
network, flat loss	$8.23 \pm 0.02\%$	$0.199 \pm 0.015\%$
ridge readout on its frozen features	$8.33 \pm 0.03\%$	$0.236 \pm 0.032\%$
Matérn head on its frozen features	$8.08 \pm 0.01\%$	$0.109 \pm 0.007\%$
network, weighted-coefficient loss	$36.46 \pm 0.65\%$	$0.0751 \pm 0.0025\%$
ridge readout on its frozen features	$40.10 \pm 0.91\%$	$0.0866 \pm 0.0039\%$
Matérn head on its frozen features	$15.47 \pm 0.47\%$	$0.0631 \pm 0.0037\%$

by hand does not change this reading. A per-dimension metric fit by the kernel-flows cross-validation loss of Owhadi and Yoo (2019) reaches the same 30% on the raw state, and a low-rank Mahalanobis metric with a thousand more parameters does not improve on it; on the features, where the network has already conditioned the representation, a learned metric gives no gain over the isotropic kernel.

5.1 What the metric buys, measured

A sweep refits both raw-input kernel rows at five nested training sizes, ten seeds per band, estimating the sensitivity map from the available training rows at each size. At $n = 18000$ the isotropic-to-adapted error ratios range from 1.1 to 1.5. The fitted log–log slopes differ little on O2 and more on strong CO_2; the numerical results are in Section S7. These slopes

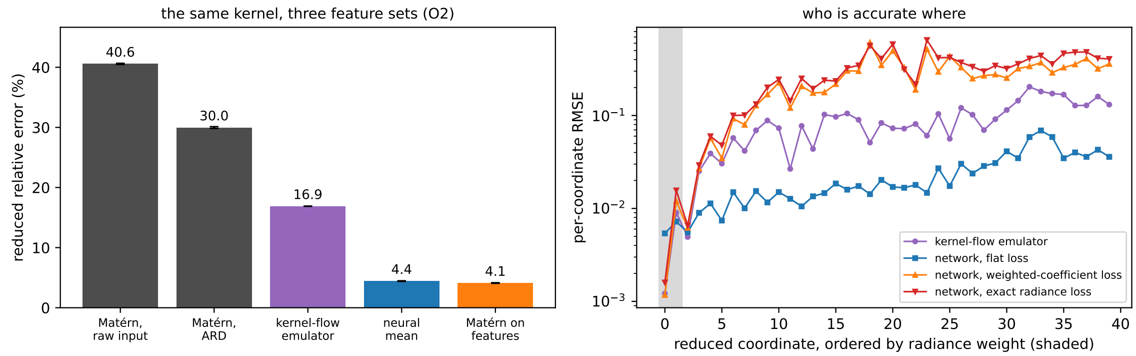

Figure 3: Both panels are drawn from the ten-seed campaign by `campaign/make_oco2_fig.py`: bars and error bars in the left panel are the mean and standard deviation over the ten seeds, curves in the right panel are per-coordinate test RMSE averaged over the same ten. Left: the representation ladder on the O2 band. The same exact Matérn machinery moves from 40% on the raw input to 4.1% on the trained network's features; the kernel-flow emulator and the network sit in between. Right: per-coordinate test RMSE with the reduced coordinates ordered by their radiance weight (shaded). Networks are named by the loss they were trained on. The per-output-tuned kernel-flow emulator is most accurate exactly on the heavily weighted leading coefficient (0.0012 against the flat network's 0.0054), while the flat network is nearly four times more accurate across the trailing thirty coordinates that the radiance barely sees (0.0276 against 0.1037). Both weighted losses buy the leading coefficient at the price of the tail, and by almost the same amount (0.0012 and 0.0016), which is the measurement behind the reading below; the per-coordinate combination takes each coordinate from whichever model wins it.

summarize a finite range of training sizes. Proposition S2.5 gives a finite-design bound for rank truncation under Lipschitz and native-space assumptions.

5.2 Transfer of the two-member rule

Part (i) of Proposition 6.1 says that a weaker member helps the optimal two-member squared-relative-error mixture exactly when $\varrho < e_1/e_2$. We examine this criterion using stored validation and test predictions for eight models: the two raw-input kernels, three networks, and three feature-kernel heads. There are $\binom{8}{2} = 28$ pairs at each of thirty band-seeds, 840 in all.

Evaluating both sides on the same block, the criterion agrees with whether the optimal convex mix beats the better member on all 840 pairs. This checks the algebraic identity against stored predictions.

The separate transfer question uses ϱ and e_1/e_2 estimated on validation, then compares that verdict with the test-block result. The test-side outcome is whether the optimal convex two-member mix beats the better of the two members *on test*, whichever that is (the order of a pair flips between the splits in 40 of the 840). The rule is correct on 62.4% of the 840 pairs. Stratified by the observed margin $|\varrho - e_1/e_2|$ that the rule must resolve, the outcome is 101 of 257 below 0.02, 95 of 215 from 0.02 to 0.05, 81 of 88 from 0.05 to 0.10, 215 of

248 from 0.10 to 0.25, and 32 of 32 above 0.25 — that is 39%, 44%, 92%, 87% and 100%. That scores the existence of a helpful mix. Scoring what the rule would ship — the mix at the weight fitted on validation, against the better test member — the counts are 142 of 257, 107 of 215, 76 of 88, 125 of 248 and 31 of 32 (57.3% overall). Those are rates at which the rule's yes-or-no prediction about the shipped outcome is right; in absolute terms 356 of the 840 transferred mixtures beat the better test member, all of them among the 715 pairs with a positive admission verdict. A verdict about the existence of a helpful mixture is different from realizing that gain using a weight transferred from validation. Every quantity here is the root-mean-square relative error and the uncentered alignment the proposition is stated in (`oco_ensemble_recheck.json` in the release). The 28 pairs of a band-seed share members and the ten seeds of a band share one test block, so the bins describe a trend across margin rather than binomial samples. What degrades in the small-margin pairs is the transfer of two estimates from one block to another, not the proposition.

The signed gap $D = \varrho - e_1/e_2$ changes between validation and test with empirical standard deviation 0.099 across these dependent pairs and mean change 0.0004. Its within-pair seed scatter is 0.010. The validation-to-test design shift is therefore material in this diagnostic; Proposition S2.1 turns such a shift into a probabilistic bound under an estimation-error condition, and the margin bins describe this pool rather than an admission threshold.

For the raw-input kernel against the flat network, the ten-seed validation means are $\varrho = 0.156$ and $e_n/e_k = 0.157$. The difference changes sign across seeds, giving an admission verdict at five of ten. The corresponding optimal validation RMS relative errors are 8.4079% for the mixture and 8.4081% for the network, at a network weight of 0.9999. The test-block hindsight optimum differs from the network by 0.0004 percentage points. These gains are too small to support a practical improvement from this pair, which accounts for its limited contribution.

5.3 Data scaling

The OCO-2 error decreases over the training sizes examined. The learning curves and the large-n ClimSim results are reported in Section S7.

5.4 Sensitivity to the kernel search grid

The boundary selections in Table 6 motivate a direct check of the search range. We train three new paired runs in each band, at seeds 0, 1, and 2, then compare two kernel grids on each fixed set of neural predictions and features. The original grid has 12 scale–nugget pairs; the expanded grid has 56. Both searches use the same validation objective, 6000-row tuning subset, and full-training refit. The raw-input, ARD-input and three feature kernels receive the same expanded candidates. Network training remains at 250 epochs. These runs form a separate sensitivity comparison alongside the ten-seed table.

Table 8 reports representative predictors under both searches. The per-seed records include both improvements and deteriorations on test. The grids are nested, but the validation winner need not improve test error. For the coordinate combination, mean reduced error rises from 4.112% to 4.329% on O2, from 16.270% to 16.393% on WCO2, and from 8.091% to 8.107% on SCO2. The radiance changes have mixed signs across seeds. The flat-feature kernel head still outperforms the raw-input kernel on the reduced criterion in

Table 8: OCO-2 search-grid sensitivity, three paired seeds per band. Entries are mean $\pm$ sample standard deviation, in percent. Each pair uses the same trained networks and validation split. The source emulator and linear-readout controls, together with every feature head and both coordinate selectors, appear in Supplement S11.

		Reduced error (%)		Radiance error (%)	
Band	Model	Recorded grid	Expanded grid	Recorded grid	Expanded grid
O2	Raw input	40.529 ± 0.089	40.529 ± 0.089	0.2353 ± 0.0047	0.2353 ± 0.0047
O2	ARD input	30.019 ± 0.035	29.062 ± 0.165	0.2706 ± 0.1030	0.1519 ± 0.0007
O2	Flat-loss network	4.440 ± 0.087	4.440 ± 0.087	0.1893 ± 0.0158	0.1893 ± 0.0158
O2	Flat-feature head	4.104 ± 0.074	4.331 ± 0.074	0.0821 ± 0.0092	0.1260 ± 0.0203
O2	Coordinate combination	4.112 ± 0.068	4.329 ± 0.075	0.0282 ± 0.0012	0.0285 ± 0.0012
WCO2	Raw input	58.625 ± 0.202	58.625 ± 0.202	0.4431 ± 0.0063	0.4431 ± 0.0063
WCO2	ARD input	52.454 ± 0.071	53.505 ± 0.073	0.2231 ± 0.0014	0.7523 ± 0.0106
WCO2	Flat-loss network	16.419 ± 0.070	16.419 ± 0.070	0.2268 ± 0.0049	0.2268 ± 0.0049
WCO2	Flat-feature head	16.270 ± 0.067	16.395 ± 0.073	0.1154 ± 0.0012	0.1770 ± 0.0046
WCO2	Coordinate combination	16.270 ± 0.068	16.393 ± 0.072	0.0537 ± 0.0069	0.0544 ± 0.0056
SCO2	Raw input	41.012 ± 0.090	43.708 ± 0.065	0.5229 ± 0.0081	0.5738 ± 0.0054
SCO2	ARD input	28.717 ± 0.021	28.717 ± 0.021	0.6023 ± 0.0097	0.6023 ± 0.0097
SCO2	Flat-loss network	8.236 ± 0.021	8.236 ± 0.021	0.2060 ± 0.0209	0.2060 ± 0.0209
SCO2	Flat-feature head	8.092 ± 0.018	8.108 ± 0.025	0.1094 ± 0.0051	0.1142 ± 0.0093
SCO2	Coordinate combination	8.091 ± 0.018	8.107 ± 0.025	0.0573 ± 0.0045	0.0566 ± 0.0033

each band. Supplement S11 reports all fourteen predictors, the paired records, boundary selections and failed-cell counts.

6 Theory

Throughout this section $\mathcal{U} = \mathbb{R}^p$ denotes the (discretized) input space and $\mathcal{V} = \mathbb{R}^q$ the output space, $G : \mathcal{U} \to \mathcal{V}$ the target operator, μ the input distribution, and $(u_1, v_1), \ldots, (u_N, v_N)$ i.i.d. draws with $v_n = G(u_n)$; the data carry no sampling noise in the usual sense, coming from a deterministic solver, though Section 4.5 discusses solver error. Assume $\|G(u)\|_2 > 0$ almost surely and that all displayed expectations are finite. We write $\ell(w, v) = \|w - v\|_2/\|v\|_2$ for the relative error and $R(\widehat{G}) = \mathbb{E}_{u\sim\mu}\, \ell(\widehat{G}(u), G(u))$ for the risk, which is the quantity reported in Section 4. The results below connect residual alignment to the scope for averaging, and kernel corrections to conditional error bounds. Most are standard identities or direct extensions of classical results; their role is to interpret the measurements in this study. The ensemble-floor, sharp kernel-bound, and minimax theorems are proved below; Supplement S8 also includes these proofs and those of the remaining results.

6.1 Guarantees for the fitted stages

Section S1 states finite-sample bounds for reflection averaging (Proposition S1.1), the kernel-flow regularizer (Proposition S1.2), and the stack, affine readout, and kernel selection stages (Propositions S1.3 and S1.5, Lemma S1.4, and Corollary S1.6). The statistical selection bounds require candidates fixed before an independent validation sample is drawn. The deployed pipeline uses validation for checkpoint, member, and kernel selection, so those bounds describe the dependence on class size, sample size, and correction weights; several are numerically vacuous at our sample sizes. Lemma S1.7 gives the uniform correction-weight bound $\kappa_{\mathrm{an}} = 500$ for the stated kernel normalization and nugget. Theorems 6.9 and 6.10 are deterministic conditional bounds, whose unknown native-space norms prevent numerical certification.

Split-conformal coverage has a separate requirement: the predictor and scale must be fixed independently of exchangeable calibration and evaluation observations.

6.2 What the stack can achieve

Proposition S1.3 bounds weight selection under independent-validation assumptions; it does not imply that mixing improves on a member. That is decided by a single measurable matrix. For a map f write the *normalized residual* at u as $\rho_f(u) = \big(f(u) - G(u)\big)/\|G(u)\|_2$, so that $\ell(f(u), G(u)) = \|\rho_f(u)\|_2$.

Proposition 6.1 (second-moment identity for convex stacks)**.** *Let $f_1, \dots, f_M$ be fixed maps and define $S \in \mathbb{R}^{M\times M}$ by $S_{mk} = \mathbb{E}_{u\sim\mu}\,\langle \rho_{f_m}(u), \rho_{f_k}(u)\rangle$. For w in the simplex Δ^{M-1} let $f_w = \sum_m w_m f_m$. Then*

$$\mathbb{E}_{u\sim\mu}\,\ell\big(f_w(u), G(u)\big)^2 \;=\; w^\top S\, w,$$

so the squared-metric risk of every convex stack is determined by S, and the best achievable is $\min_{w\in\Delta^{M-1}} w^\top S w$. In particular:

- *(i) (two members) if S has diagonal (e_1^2, e_2^2) with $0 < e_1 \le e_2$ and uncentered correlation $\varrho = S_{12}/(e_1 e_2)$, mixing in the weaker member strictly helps if and only if $\varrho < e_1/e_2$, in which case the optimum is interior with value $e_1^2 e_2^2(1-\varrho^2)/(e_1^2 + e_2^2 - 2\varrho e_1 e_2)$;*
- *(ii) (equicorrelated family) if $S_{mm} = e^2$ and $S_{mk} = \varrho\, e^2$ for all $m \ne k$ with $\varrho \ge 0$, then $\min_w w^\top S w = e^2\big(\varrho + (1-\varrho)/M\big)$, attained at uniform weights and decreasing to $e^2\varrho$ as $M \to \infty$.*

Proof. See Supplement S8.10.

If a member has zero squared risk, it already attains the minimum zero; ratios involving its error are not used.

Part (i) is the classical two-member minimum-variance rule, and part (ii) is the equicorrelated case of the ambiguity decomposition of ensemble learning (Krogh and Vedelsby, 1995); see Ueda and Nakano (1996) for the generalization-error form and Brown et al. (2005) for a survey. Both enter here as measurable decision tools on the estimated S.

Everything in this subsection is stated for the squared metric, and the tables of Sections 4 and 5 report a different one. The distinction is small numerically and decisive logically, so we make it explicit once and refer back to it wherever the two families of numbers meet.

Remark 6.2 (the two error metrics)**.** *For a map f write $\mathcal{E}_1(f) = \mathbb{E}\|\rho_f(u)\|_2$ and $\mathcal{E}_2(f) = (\mathbb{E}\|\rho_f(u)\|_2^2)^{1/2}$, the mean and the root mean square of the per-sample relative error. If $\mathcal{E}_1 = 0$, both errors vanish and no ratio is needed; below assume $\mathcal{E}_1 > 0$. The tables report*

$\mathcal{E}_1$; Proposition 6.1 is a statement about $\mathcal{E}_2$, since $\mathcal{E}_2(f_w)^2 = w^\top S w$ by definition. Writing $X = \|\rho_f(u)\|_2$ and $c_f = (\operatorname{Var} X)^{1/2}/\mathbb{E}X$,

$$\mathcal{E}_2(f)^2 - \mathcal{E}_1(f)^2 = \operatorname{Var} X, \qquad \mathcal{E}_1(f)/\mathcal{E}_2(f) = (1 + c_f^2)^{-1/2} \le 1,$$

with equality only if the per-sample error is almost surely constant. Two consequences are used in this paper. The floor belongs in $\mathcal{E}_2$, because only $\mathcal{E}_2$ is determined by S: the map $w \mapsto w^\top S w$ is a quadratic form in second moments, whereas $w \mapsto \mathbb{E}\|\sum_m w_m \rho_{f_m}(u)\|_2$ depends on the full joint law of the residuals. Upper bounds transfer downward, since $\mathcal{E}_1 \le \mathcal{E}_2$, while lower bounds do not — so a floor read off S says nothing about a table entry without the dispersion of the same predictor. For the structural-mechanics stack the mean is about six percent below the RMS; for the O2 predictors of Section 5, the difference is ten to eleven percent. The dispersion is a property of the predictor and not a constant of the problem, so a single conversion factor should not be reused across objects.

Applying part (i) requires estimating the errors and their correlation. The reliability of that decision depends on its distance from the threshold relative to the estimation error. A margin law for threshold rules of this shape (Proposition S2.1, Supplement S2) bounds the probability that a decision is both wrong and taken at an observed margin of at least τ by $\exp(-\tau^2/2\sigma^2)$, with σ the scale of the estimation error and not a chosen constant. That is a joint bound, not the error rate among the decisions whose observed margin exceeds τ: the conditional quantity is the joint one divided by $\mathbb{P}(|\widehat{D}| \ge \tau)$, on which the proposition says nothing. Accuracy inside a margin bin is therefore an empirical calibration of the rule, and Section 5.2 reports it that way.

The equicorrelated model is restrictive. A lower bound for convex weights needs only one-sided bounds on the entries of S.

Corollary 6.3 (ensembling floor)**.** *If every member satisfies $S_{mm} \ge \bar{e}^{\,2}$ and every pair satisfies $S_{mk} \ge \bar{\varrho}\,\bar{e}^{\,2}$ with $\bar{\varrho} \in [0,1]$, then every convex combination f_w obeys*

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 \;=\; w^\top S w \;\ge\; \bar{e}^{\,2}\big(\bar{\varrho} + (1-\bar{\varrho})\,\|w\|_2^2\big) \;\ge\; \bar{e}^{\,2}\,\bar{\varrho}.$$

No number of additional members with the same error level and mutual correlations moves the ensemble's root-mean-square relative error below $\bar{e}\sqrt{\bar{\varrho}}$.

Proof. See Supplement S8.11.

Upper bounds on the same entries give an upper bound on the optimum by evaluating uniform weights.

Corollary 6.4 (the floor is nearly achieved)**.** *Suppose $\bar{e} > 0$ and, in addition to the hypotheses of Corollary 6.3, that $S_{mm} \le \bar{e}^{\,2}(1+\delta_e)$ for every m and $S_{mk} \le (\bar{\varrho} + \delta_\varrho)\,\bar{e}^{\,2}$ for every $m \ne k$. Then*

$$\bar{\varrho} \;\le\; \frac{1}{\bar{e}^{\,2}} \min_{w \in \Delta^{M-1}} w^\top S w \;\le\; \bar{\varrho} + \delta_\varrho + \frac{1 + \delta_e - \bar{\varrho} - \delta_\varrho}{M}.$$

The width of the bracket is the normalized off-diagonal entry spread δ_ϱ plus a $1/M$ term: when these entrywise spreads are small, the optimal convex risk is localized. The upper bound is obtained by evaluating uniform weights; it does not upper-bound an arbitrary fitted convex combination.

Proof. See Supplement S8.6.

Applied to a measured matrix $\widehat{S}$, these are exact empirical bounds on that same design, not confidence bounds on the population matrix. For the six-member validation matrices, the weak RMS floor $\sqrt{\min_{i\neq j}\widehat{S}_{ij}}$ is $4.820 \pm 0.062\%$, the stronger finite-six-member lower bound is $4.849 \pm 0.060\%$, and the simplex optimum is $4.920 \pm 0.057\%$ (mean and sample standard deviation over ten seeds). The values near 4.91% obtained from mean centered correlations are equicorrelation diagnostics, not these lower bounds.

The floor is stated for convex weights, and the deployed stack is not convex: its per-pixel weights are affine and unconstrained in sign. For global weights the sign constraint can be removed exactly.

Corollary 6.5 (signed stacks)**.** *If S is positive definite, the minimum of $w^\top Sw$ over all w with $\sum_m w_m = 1$, of either sign, is $1/(\mathbf{1}^\top S^{-1}\mathbf{1})$, attained at $w^\star = S^{-1}\mathbf{1}/(\mathbf{1}^\top S^{-1}\mathbf{1})$; it is at most the convex minimum, with equality exactly when $w^\star$ has no negative coordinate.*

Proof. See Supplement S8.9.

On the measured S the two minima differ by at most 0.00052 percentage points at every seed and for both member sets, but not because the constraint is slack: the unconstrained minimizer carries a small negative coordinate at seven seeds of ten for the six members and at one seed for the five (the stored `signed` records). The effect of allowing signed global weights is therefore small here. What the deployed per-pixel stack gains over the convex one (0.032 and 0.035 points in the two campaigns) is a benefit of a different per-pixel affine class; the global signed-weight comparison does not isolate the effect of signs or intercepts within that class.

The members of a pool are usually of two kinds at once, architectures and seeds of each architecture, and the floor separates the two.

Theorem 6.6 (seeds and architectures)**.** *Let the pool consist of K seeds of each of M architectures, and suppose $S_{ii} \geq \bar{e}^2$ for every member, $S_{ij} \geq \varrho_w \bar{e}^2$ for two seeds of the same architecture and $S_{ij} \geq \varrho_b \bar{e}^2$ for members of different architectures, with $0 \leq \varrho_b \leq \varrho_w \leq 1$. Then every convex combination obeys*

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 \;\geq\; \bar{e}^2\Big(\varrho_b + \frac{\varrho_w - \varrho_b}{M} + \frac{1-\varrho_w}{MK}\Big),$$

with equality at uniform weights when the three inequalities are equalities. For an extended pool only if the same entrywise bounds continue to hold, no number of seeds of one architecture takes the root-mean-square error below $\bar{e}\sqrt{\varrho_w}$, no number of seeds of M architectures takes it below $\bar{e}\sqrt{\varrho_b + (\varrho_w - \varrho_b)/M}$, and no pool of any size takes it below $\bar{e}\sqrt{\varrho_b}$.

Proof Let $W_a = \sum_{k=1}^{K} w_{a,k}$ be the weight assigned to architecture a. Then $\sum_a W_a = 1$. Nonnegativity of the weights lets us apply the three entrywise lower bounds separately:

$$\begin{aligned} w^\top Sw &\geq \bar{e}^2\left[\|w\|_2^2 + \varrho_w\left(\sum_a W_a^2 - \|w\|_2^2\right) + \varrho_b\left(1 - \sum_a W_a^2\right)\right] \\ &= \bar{e}^2\left[\varrho_b + (\varrho_w - \varrho_b)\sum_a W_a^2 + (1-\varrho_w)\|w\|_2^2\right]. \end{aligned}$$

Cauchy–Schwarz gives $\sum_a W_a^2 \geq 1/M$ and $\|w\|_2^2 \geq 1/(MK)$. Their coefficients are non-negative, so substitution proves the bound. Uniform weights make both Cauchy–Schwarz inequalities equalities. When the entrywise bounds are equalities too, every step is an equality. The stated limits follow directly, provided the same bounds hold throughout the enlarged pool. ■

At fixed M, increasing K reduces only the term $(1-\varrho_w)/(MK)$ in the lower bound. The within-architecture contribution $(\varrho_w - \varrho_b)/M$ persists. Section 4.6 applies the formula empirically to sixty trained predictors; it does not assume the same inequalities for untrained members.

Differentiating the two-member optimum in Proposition 6.1(i) quantifies the local trade-off between accuracy and correlation.

Proposition 6.7 (the price of accuracy in correlation)**.** *In the setting of Proposition 6.1(i) write $t = e_2/e_1$ and assume $-1 < \varrho < 1/t$ (with $t \geq 1$), so the optimum is interior. The optimal two-member risk is $V = e_1^2 t^2(1-\varrho^2)/(1+t^2-2\varrho t)$, and*

$$\frac{\partial V}{\partial t} = \frac{2e_1^2\, t\,(1-\varrho^2)(1-\varrho t)}{(1+t^2-2\varrho t)^2} > 0, \qquad \frac{\partial V}{\partial \varrho} = \frac{2e_1^2\, t^2(t-\varrho)(1-\varrho t)}{(1+t^2-2\varrho t)^2} > 0.$$

A change $(\mathrm{d}t, \mathrm{d}\varrho)$ *in the second member therefore leaves the optimal risk unchanged to first order exactly when* $\mathrm{d}\varrho = -(1-\varrho^2)\,\mathrm{d}t/\big(t(t-\varrho)\big)$*: an improvement of the second member's error by a fraction ϵ is cancelled by a rise of $\epsilon(1-\varrho^2)/(t-\varrho)$ in its correlation with the first.*

Proof. See Supplement S8.13.

At perfect anticorrelation $\varrho = -1$, cancellation gives zero optimal risk and the derivative in t is zero, so that boundary is excluded above. At the admission threshold $\varrho = 1/t$ with $t > 1$, the exchange coefficient $(1-\varrho^2)/(t-\varrho)$ tends to $1/t$; it need not be large. The formula is a local sensitivity calculation at an interior point. Supplement S6 evaluates it at the measured values.

For coordinate-dependent combinations, squared losses with a fixed diagonal metric separate by coordinate (Proposition S2.2; proof in Supplement S8). The same holds after a fixed sample weighting (Corollary S2.3; proof in Supplement S8). These statements motivate the selector but do not make it simultaneously optimal for the two reported mean relative norms: the square root couples coordinates, and the two criteria also have different sample denominators. Proposition S2.4 gives an independent-validation selection bound (proof in Supplement S8); the deployed candidates reuse validation, as noted above. The OCO-2 combination improves on the rescored source emulator in both mean criteria, while the flat feature head is slightly better on the O2 reduced criterion.

6.3 Metric and representation: when a fixed kernel loses to a network

On structural mechanics, the neural mean and the isotropic kernel differ by less than half a percentage point. On the O2 band, raw-input kernel error is roughly nine times the network's error. We examine input scaling and learned features as two possible explanations for this difference.

The first is input geometry. A sensitivity-based linear transformation improves the raw-input kernel by a factor of about 1.1–1.5 in this study. Proposition S2.5 in Supplement S2 gives a finite-design bound for a Lipschitz target after a rank-r truncation, with its approximation error explicit; its proof is in Supplement S8. It makes no assertion that a measured approximate rank changes an asymptotic rate.

The second is the representation used by the kernel. Native-space bounds depend on the target's norm in that space. A fixed kernel may approximate targets outside its RKHS, so native-space membership is not a limit on approximation expressivity. Learned features can change the relevant norm and design geometry, but neither change is automatically favorable.

The representational term has a precise reading through the same optimal-recovery bound that Section 6.4 makes exact. That bound controls the error by $\|G - m\|_{\mathcal{H}} \cdot P_\lambda$, a product of the target's norm in the kernel's native space and a design factor determined by the kernel, training inputs, and query. Changing the features changes both in principle, and on the O2 band the two move by very different amounts. The effective dimension of the Matérn Gram, the quantity Lemma S5.1 controls, is nearly identical on the raw input and on the learned features (at $n = 4000$ and a matched median length scale, 3995 against 3993 at a 10^{-8} nugget); the design factor itself is not fixed by that trace, and measured directly at the 2000 test inputs the feature kernel's power function is smaller at the median scale, its median 1.5 times below the raw kernel's, and equal to it (0.99) at the multiplier 4 that the campaign selected for every feature head. The fitted-norm diagnostic moves far more: regressing the same outputs through the two kernels, the squared native-space norm of the kernel fit, $\mathrm{tr}(\alpha^\top K\alpha)$, is 38 times smaller on the features than on the raw input at the median scale (77 times on the full 18000-row block) and 273 times smaller at the selected multiplier (139 times on the full block). We use $\mathrm{tr}(\alpha^\top K\alpha)$ rather than $\mathrm{tr}(\alpha^\top Y)$, which also includes $n\lambda\|\alpha\|_F^2$. These ratios are finite-design diagnostics of the interpolating regime: for a target in the corresponding RKHS with compatible labels, the fitted norm is a lower bound on its norm, the ratio grows with the kernel scale — 6 at half the median length scale, 154 at twice it — and at a 10^{-4} nugget, where the ridge no longer interpolates, it collapses and at the selected multiplier inverts (0.2). Both designs are standardized by their training moments before the kernel is formed, as the campaign's heads are, and the check is at one seed. Read as diagnostics, they say that the learned features hand the same kernel a target it fits with a much smaller fitted RKHS norm, and a somewhat smaller pointwise design factor at the median scale; they do not identify a causal decomposition into a fixed design factor and a moving norm.

The associated feature-kernel function space has a standard pullback description. Writing φ for the feature map and k for the base kernel, the deep kernel head uses $k_\varphi(x, x') = k(\varphi(x), \varphi(x'))$, the construction studied as deep kernel learning (Wilson et al., 2016; Calandra et al., 2016).

Proposition 6.8 (feature pullback)**.** *The native space of k_φ is $\mathcal{H}_{k_\varphi} = \{f \circ \varphi : f \in \mathcal{H}_k\}$ with $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : f \circ \varphi = g \text{ on } \mathcal{U}\}$. In particular, if the target factors through the features as $G = h \circ \varphi$ with $h \in \mathcal{H}_k$, then $\|G\|_{\mathcal{H}_{k_\varphi}} \le \|h\|_{\mathcal{H}_k}$, and the optimal-recovery bound of Theorem 6.9 for the feature-kernel regression is governed by $\|h\|_{\mathcal{H}_k}$ rather than by the norm of G in the raw-input space.*

Proof. See Supplement S8.17.

The statement is for scalar f; for the vector-valued targets of the campaigns it is applied coordinatewise, with the norm the root sum of squares over output coordinates, which is the norm Theorem 6.9 uses.

The proof (Supplement S8) is the standard pullback identity for reproducing kernels (Saitoh, 1997; Paulsen and Raghupathi, 2016). The identity specifies the feature-kernel RKHS and the minimum extension norm. It provides no comparison with the raw-input RKHS without additional assumptions. In particular, a linear feature map is itself an input metric transformation. The measured fitted-norm reduction is evidence about these fitted kernels at the reported scales and nuggets, rather than a theorem about representation learning.

6.4 An error bound for the residual kernel correction

The final stage of the pipeline is kernel ridge regression of the ensemble residual. The bound below is a vector-valued, residual form of the classical power-function estimate for kernel interpolation (Wendland, 2004) and of the optimal-recovery viewpoint of Owhadi and Scovel (2019); we include the short argument in Supplement S8 to keep the two factors explicit. Fix the mean $m : \mathcal{U} \to \mathcal{V}$ (in the pipeline, the stacked ensemble; in the statement below m is any realized map; no sample independence is needed for this deterministic statement) and write $r = G - m$ for the residual operator with components r_j, $j = 1, \ldots, q$. Let k be a positive definite kernel on $\mathcal{U}$ with RKHS $\mathcal{H}_k$, and assume $r_j \in \mathcal{H}_k$ for all j with

$$\|r\|_K^2 := \sum_{j=1}^{q} \|r_j\|_{\mathcal{H}_k}^2 < \infty.$$

Given inputs $X = (u_1, \ldots, u_n)$, Gram matrix $K = k(X, X)$ and nugget $\lambda \geq 0$, the correction is $\widehat{r}_\lambda(u) = k(u, X)\,(K + n\lambda I)^{-1} R$, $R_{nj} = r_j(u_n)$, and the corrected predictor is $m + \widehat{r}_\lambda$. Let

$$P_\lambda(u)^2 \;=\; k(u, u) - k(u, X)\,(K + n\lambda I)^{-1} k(X, u) \tag{4}$$

denote the Gaussian process posterior variance with the same nugget (for $\lambda = 0$ this is the classical power function of the point set X).

Theorem 6.9 (conditional power-function bound)**.** *For every $u \in \mathcal{U}$ and every $\lambda \geq 0$ (for $\lambda = 0$ take K positive definite, or read K^{-1} as the pseudoinverse),*

$$\big\|G(u) - m(u) - \widehat{r}_\lambda(u)\big\|_2 \;\leq\; \|G - m\|_K\, \widetilde{P}_\lambda(u) \;\leq\; \|G - m\|_K\, P_\lambda(u),$$

where $\widetilde{P}_\lambda(u)^2 \;=\; P_\lambda(u)^2 - n\lambda\, \|(K + n\lambda I)^{-1} k(X, u)\|_2^2$. The first inequality is sharp: for every u there is a residual operator of norm $\|G - m\|_K$ at which it holds with equality, so $\|G - m\|_K \widetilde{P}_\lambda(u)$ is the exact worst-case error of the correction over the ball $\{r : \|r\|_K \leq \|G - m\|_K\}$.

Proof Put $c = (K + n\lambda I)^{-1} k(X, u)$ and $g_u = k(u, \cdot) - \sum_i c_i k(u_i, \cdot)$. The reproducing property expresses each component of the error as $r_j(u) - \widehat{r}_{\lambda,j}(u) = \langle r_j, g_u \rangle_{\mathcal{H}_k}$. Consequently,

$$\|r(u) - \widehat{r}_\lambda(u)\|_2^2 \leq \sum_j \|r_j\|_{\mathcal{H}_k}^2 \|g_u\|_{\mathcal{H}_k}^2 = \|r\|_K^2 \|g_u\|_{\mathcal{H}_k}^2.$$

Since $Kc = k(X,u) - n\lambda c$, expansion of the squared norm gives

$$\|g_u\|_{\mathcal{H}_k}^2 = k(u,u) - 2c^\top k(X,u) + c^\top Kc = P_\lambda(u)^2 - n\lambda\|c\|_2^2.$$

This proves both inequalities. If $\|g_u\| > 0$, take $r_1 = \rho g_u/\|g_u\|$ and $r_j = 0$ for $j > 1$, for any prescribed radius ρ. Its norm is ρ and its error is exactly $\rho\|g_u\|$, which proves sharpness. If $g_u = 0$, every residual has zero error. For $\lambda = 0$ and singular K, take $z \in \ker K$. The function $h_z = \sum_i z_i k(u_i, \cdot)$ has squared norm $z^\top Kz = 0$, so $z^\top k(X,u) = \langle h_z, k(u,\cdot)\rangle = 0$. Thus $k(X,u)$ is orthogonal to $\ker K$ and belongs to $\operatorname{range} K$. Consequently $c = K^+ k(X,u)$ satisfies $Kc = k(X,u)$, and the same argument applies. ∎

The correction labels. Let $M_{\text{tr}} \in \mathbb{R}^{n\times q}$ denote the training-row mean actually used by the implementation, and let m_{full} be the query-time mean. The kernel channel uses foldwise fits with pooled target centering; neural channels are in-sample, and the refiner's kernel input also changes at deployment. Thus M_{tr} need not be the restriction of a single cross-fitted function. Define

$$c_\lambda(u) = (K + n\lambda I)^{-1} k(X,u), \qquad \Delta = m_{\text{full}}(X) - M_{\text{tr}}, \qquad R_{\text{tr}} = G(X) - M_{\text{tr}}.$$

The implemented predictor is $m_{\text{full}}(u) + c_\lambda(u)^\top R_{\text{tr}}$. If $G - m_{\text{full}} \in H_k^q$, Theorem 6.9 and the triangle inequality give

$$\begin{gathered}\|G(u) - m_{\text{full}}(u) - c_\lambda(u)^\top R_{\text{tr}}\|_2 \le \|G - m_{\text{full}}\|_K \widetilde{P}_\lambda(u) + \|c_\lambda(u)^\top \Delta\|_2, \\ \|c_\lambda(u)^\top \Delta\|_2 \le \|c_\lambda(u)\|_2 \|\Delta\|_F.\end{gathered} \tag{5}$$

Indeed, $R_{\text{tr}} = (G - m_{\text{full}})(X) + \Delta$; substituting this identity and applying the theorem proves the display. Section 4.8 measures the propagated mismatch on the test cases. The first term still contains an unknown native-space norm, so the display is not an evaluated error certificate for the reported correction.

The RKHS membership and the norm of the target residual are not observed. For labels sampled from one compatible $r \in H_k^q$, ridge contraction gives $\|\widehat{r}_\lambda\|_K \le \|r\|_K$. With the mixed training labels, the fitted norm need not lower-bound $\|G - m_{\text{full}}\|_K$. Table S6.2 therefore describes fitted functions only: its squared-norm ratio is about 4900 before energy normalization and 8.8 after it. Neither ratio proves a corresponding reduction in an unknown residual norm or smoothness. The design factor is a worst-case error factor and need not rank realized errors well; Section 4.7 measures that ranking directly.

The interpolant also minimizes worst-case error among all rules using the same residual observations. The next result states that classical recovery property and gives the increase in the worst-case constant caused by ridge regularization.

Theorem 6.10 (minimax optimality over the native-space ball)**.** *Fix a design X with K positive definite, a point u and a radius $\rho > 0$. For every map $\Psi : \mathbb{R}^{n\times q} \to \mathbb{R}^q$ that estimates $r(u)$ from the training residuals $r(X)$,*

$$\sup_{\|r\|_K \le \rho} \big\|r(u) - \Psi\big(r(X)\big)\big\|_2 \;\ge\; \rho\, P_0(u),$$

and the kernel interpolant $\Psi(R) = k(u,X)K^{-1}R$ attains the bound. For $\lambda > 0$ the worst case of the ridge correction over the same ball is $\rho\,\widetilde{P}_\lambda(u)$, and, with $k_u = k(X,u)$ and $A_\lambda = K + n\lambda I$,

$$P_0(u)^2 \;\le\; \widetilde{P}_\lambda(u)^2 \;\le\; P_\lambda(u)^2, \qquad \widetilde{P}_\lambda(u)^2 - P_0(u)^2 = (n\lambda)^2\, k_u^\top A_\lambda^{-2} K^{-1} k_u,$$

so the nugget's cost in the worst-case constant is a computable quantity that vanishes as $\lambda \to 0$.

Proof At $\lambda = 0$, the function g_u in the preceding proof vanishes on X, has norm $P_0(u)$, and satisfies $g_u(u) = P_0(u)^2$. If $P_0(u) > 0$, the two residuals $r^\pm = \pm\rho g_u e_1/P_0(u)$ belong to the ball and give identical observations, namely zero. Any rule therefore returns the same vector $\psi = \Psi(0)$ for both. The triangle inequality gives

$$2\rho P_0(u) = \|r^+(u) - r^-(u)\|_2 \le \|r^+(u) - \psi\|_2 + \|r^-(u) - \psi\|_2,$$

so at least one error is $\rho P_0(u)$ or larger. When $P_0(u) = 0$ this lower bound is immediate. The preceding theorem supplies the matching upper bound for interpolation and the exact ridge worst case. Finally, set $t = n\lambda$ and $A = K + tI$. Because A commutes with K,

$$\begin{aligned}\widetilde{P}_\lambda^2 - P_0^2 &= k_u^\top (K^{-1} - 2A^{-1} + A^{-1}KA^{-1}) k_u \\ &= k_u^\top A^{-2} K^{-1} (A-K)^2 k_u = t^2 k_u^\top A^{-2} K^{-1} k_u \ge 0.\end{aligned}$$

The upper inequality follows from the preceding theorem. For the fixed finite positive definite matrix K, the last expression tends to zero as $t \to 0$. ■

The lower bound is the classical optimal-recovery argument (Owhadi and Scovel, 2019): two residuals of norm ρ that vanish on the design and take opposite values at u are indistinguishable from the data, so any rule errs by at least $\rho P_0(u)$ on one of them. What the theorem adds for the pipeline is a reading of the reported quantity: at $\lambda = 0$ the worst-case constant $\|G - m\|_K\, P_0(u)$ over the native-space ball is the smallest any rule can attain from these data, and at the validated nugget the gap to it is the displayed correction term, evaluable from the Gram matrix alone. The statement is about the worst case over a ball whose radius is not known; it is not an operational error bar for the fitted surrogate, which is the split-conformal set of the next subsection.

6.5 Coverage of calibrated prediction balls

Theorem 6.9 controls the error by $\|G - m\|_K\, P_\lambda(u)$, but the constant $\|G - m\|_K$ is not known a priori and Section 4.7 shows that P_λ alone ranks the error weakly. What the surrogate reports as uncertainty is therefore a rescaling $Q\, P_\lambda(u)$, whose multiplier is the split-conformal quantile of the scores $\|e(\tilde{u}_i)\|_2 / P_\lambda(\tilde{u}_i)$ on a calibration split of the test block, drawn after every selection in the pipeline has been made on the validation split; Section 4.7 reports this band beside a constant-radius band calibrated the same way from the raw scores $\|e(\tilde{u}_i)\|_2$. Coverage does not require the RKHS bound or a useful error ranking. For a predictor and positive scale fixed independently of exchangeable calibration and future inputs, Proposition S3.1 gives marginal coverage at least $1 - \alpha$. Its proof is in Supplement S8.

The upper bound $1 - \alpha + 1/(m+1)$ additionally requires almost surely distinct scores. Define the conformal quantile using the $k = \lceil (1-\alpha)(m+1) \rceil$-th order statistic of the m scores augmented by $+\infty$, which also covers $k = m + 1$.

If, conditional on the fitted predictor and scale, the calibration and evaluation scores are i.i.d. with a continuous distribution and $k \leq m$, the random conditional coverage over calibration draws has law $\mathrm{Beta}(k, m+1-k)$. The number covered among N fresh evaluation cases is Beta–binomial. Remark S3.2 states and derives this law in Supplement S3. At $m = 1000$ and $N = 19000$ its nominal-90% central reference interval is $[88.0\%, 91.8\%]$. The coverage experiment is compared with this reference law under those conditions; it is not a guarantee conditional on each input or on arbitrary prior test-driven selection.

7 Discussion

The two benchmarks give different roles to a kernel fitted after neural training. On structural mechanics, the neural members and their combination account for most of the accuracy, with a smaller repeatable gain from residual correction. On OCO-2, the representation learned by the network is the main advantage: the direct-target kernel head performs substantially better on frozen features than on raw inputs. Coordinate selection then balances errors across two metrics that weight the reduced outputs very differently.

The structural-mechanics score of $4.546 \pm 0.003\%$ is close to PARA-Net's published 4.55%; the two studies differ in training subset and seed protocol, so the comparison is indicative rather than a ranking. The improvement in the lower-data regime is larger. The paired reruns in Section 4.8 change target centering while repeating downstream training and fitting. Their mean change is only 0.000011 percentage points, with changes in both directions across seeds. Replacing correction labels by residuals of the query-time mean also has a small effect under the fixed kernel. These checks quantify two implementation choices on this benchmark.

The ensemble analysis gives a more specific conclusion. For the sixty trained predictors, the empirical lower bound of 4.814% lies close to the hindsight-optimal global convex RMS error of 4.876%. There is little room left for this form of averaging within the pool. Both individual accuracy and residual alignment matter: completing the FNO schedule improves its own error without improving the five-member pipeline, whereas adding the UNet improves the corrected pipeline by 0.032 percentage points. That last result comes from an ablation of the full estimator, whose per-pixel weights and kernel correction extend beyond the class covered by the convex bound; the shared residual is a property of these predictors rather than an irreducible floor for the benchmark.

The OCO-2 comparison benefits from access to the source emulator's predictions on the same test states. Its scope is the retained forty-component representation: full-spectrum truncation, instrument response, Jacobians, and retrieval accuracy are not evaluated. The coordinate-selected predictor stays close to the best head on the reduced criterion while improving radiance error, rather than dominating every constituent on both. A frozen linear-readout control supports the benefit of the nonlinear kernel head in these experiments. The wider matched kernel search in Section 5.4 preserves the feature advantage over raw-input kernels, but its validation-selected coordinate combination has higher mean reduced error in all three bands. A larger candidate set does not guarantee lower test error.

The theory helps interpret these results. The native-space bound separates a residual norm from a design factor and gives the exact worst-case cost of a positive ridge parameter. Applying it to the implemented correction also requires the label-mismatch term in Section 6.4. The bound is conditional on the residual's RKHS membership and norm, which are not assumed for the benchmark target. Fitted norms describe the finite-design regressions. The pullback identity characterizes the space induced by neural features; whether its norm is smaller than the raw-input norm is a property of the fitted features rather than of the identity.

For uncertainty quantification, the simpler scale works better here. Both kernel power functions and ensemble disagreement rank errors weakly, and a constant-radius calibrated set is narrower on average. All three constructions carry the same marginal split-conformal guarantee under the exchangeability conditions of Section 6.5.

The supplement provides the proofs, the paired sensitivity results, and the computational details behind the reported numbers.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital; both entities supported this work and are named above. Neither had a role in the design of the study, the analysis, the writing, or the decision to publish. The public funders had no role in those either.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The author made substantial use of Claude Code (Anthropic) and Codex (OpenAI) for software development, numerical diagnostics, mathematical calculations, literature checks, and drafting and editing the manuscript. This assistance included benchmark preprocessing and adaptation of the publicly released OCO-2 emulator code into the Python experiment pipeline. Separate AI-agent reviews examined mathematical arguments, implementation, and exposition as internal checks. The author is responsible for the analysis, citations, conclusions, and all other content of the manuscript.

Data and code availability

The code, the per-run summaries behind every reported number, and the exact configuration of each experiment are available at `https://github.com/yspennstate/neural-means-kernel-corrections`. The repository's `paper/` directory contains the article, the supplement, and their sources. Full checkpoints and large prediction/residual arrays are not bundled in the repository and are available from the author on request. The structural-mechanics, Helmholtz, Navier–Stokes and advection data are distributed through the data record accompanying de Hoop et al. (2022) (`data.caltech.edu`, record `20091`); the OCO-2 emulation data and the kernel-flow emulator's predictions through the OSF project `u2t8a`; and the ClimSim data through the LEAP repository `subsampled_low_res`.

Supplementary material

The supplement, referenced throughout as Supplement S1–S11, is provided as the separate file `supplement.pdf` and is part of this manuscript. The accompanying versioned source package contains this article, its matching supplement, and their editable sources. The three main theorems are proved in the article; the supplement collects every proof, and each formal result identifies its specific proof location. It contains the finite-sample bounds for every fitted stage and all proofs, the margin and per-coordinate selection results, the effective-dimension identity, the ablation, error-localization, spectra, uncertainty-quantification and cost detail on structural mechanics, the uncontrolled survey of the remaining suite problems and the mirror-symmetry check, the OCO-2 input-metric study with the data-scaling and ClimSim results, the implementation and compute record, and the numerical checks of the stated identities and inequalities, and the paired centering, correction-label, and kernel-grid sensitivity experiments. Both documents and their editable sources are also included in the accompanying Overleaf package.

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